

PrivML: Test Outputs

Timestamp: 2025-09-19 19:25:46

Model ID: Model 51

Dataset Path: data/Humans

Number of Images in Dataset: 3000

Training Size: 75%

Validation Size: 15%

Testing Size: 5%

Encrypted Testing Subset Size: 5%

Random Seed: 500

Plain Model Training

Epochs: 150

Batch Size: 16

Learning Rate: 0.001

Optimizer: Adam

Loss Function: MSELoss

Sample Output Count: 3

Plain Model Architecture

Conv1 Kernel Size: 3

Conv2 Kernel Size: 1

Conv1 Padding: 1

FHE Configuration

Encryption Scheme: CKKS

Scale (bits): 40

Polynomial Modulus Degree: 32768

Key Generation Level: 60

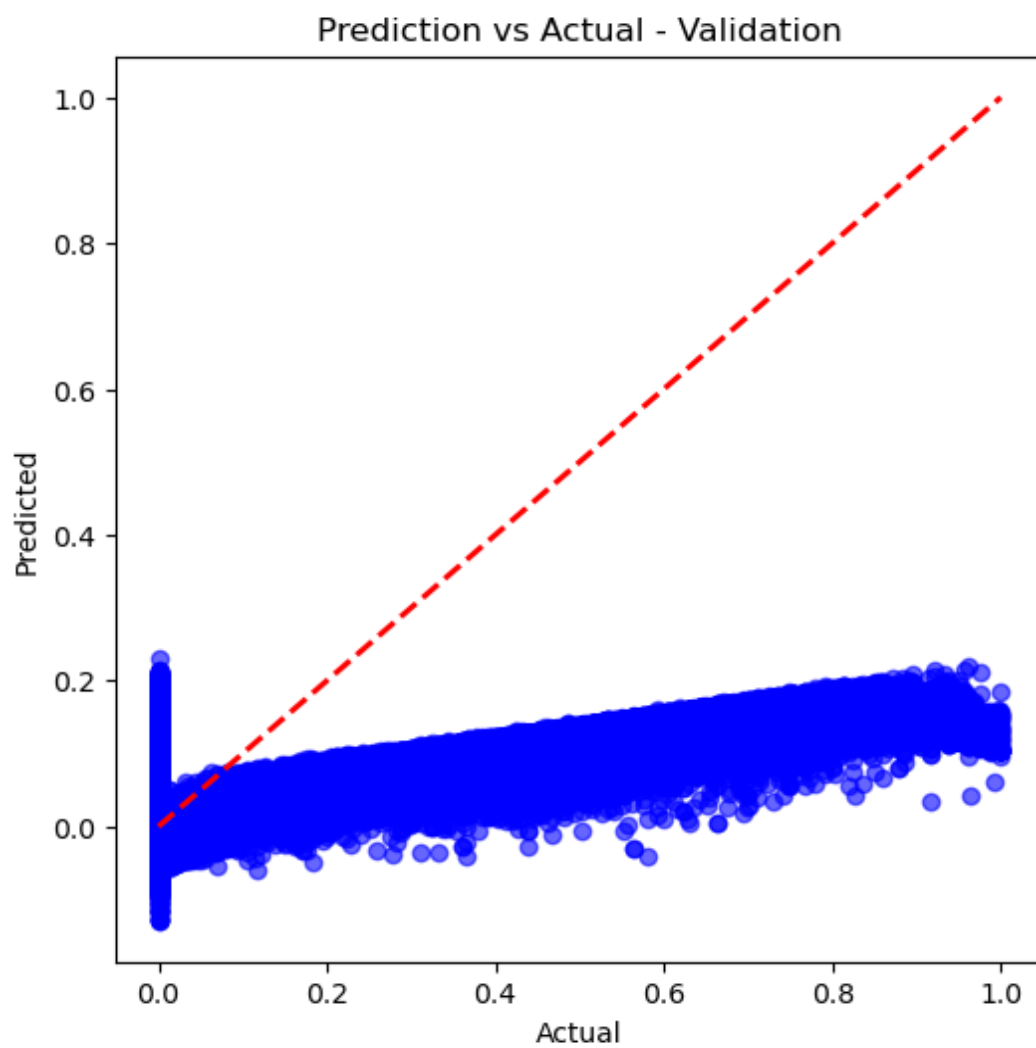
Decryption Stability Level: 60

Encrypted Loss Function: MSELoss

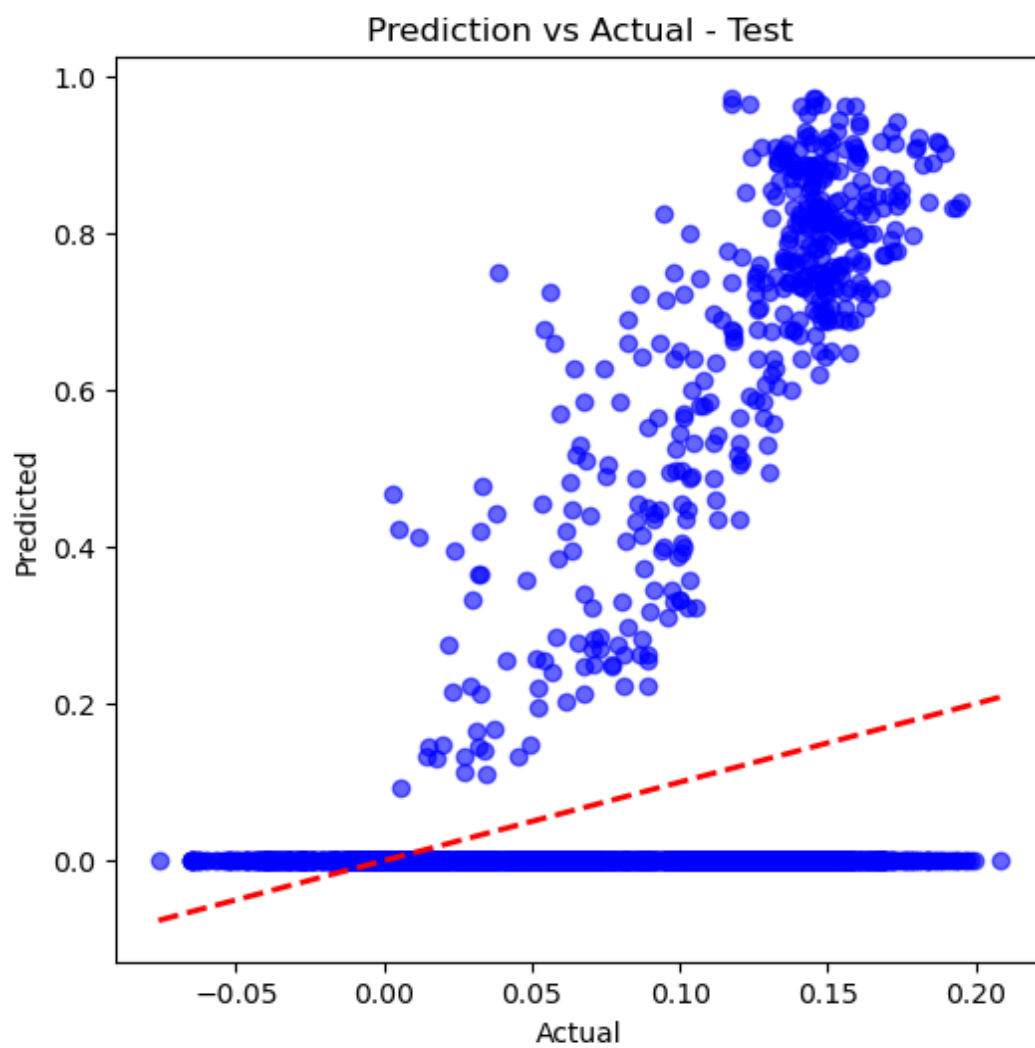
Coefficient Mod Bit Sizes: [60, 40, 40, 40, 40, 60]

Privacy-Preserving ML: ML Metrics Report

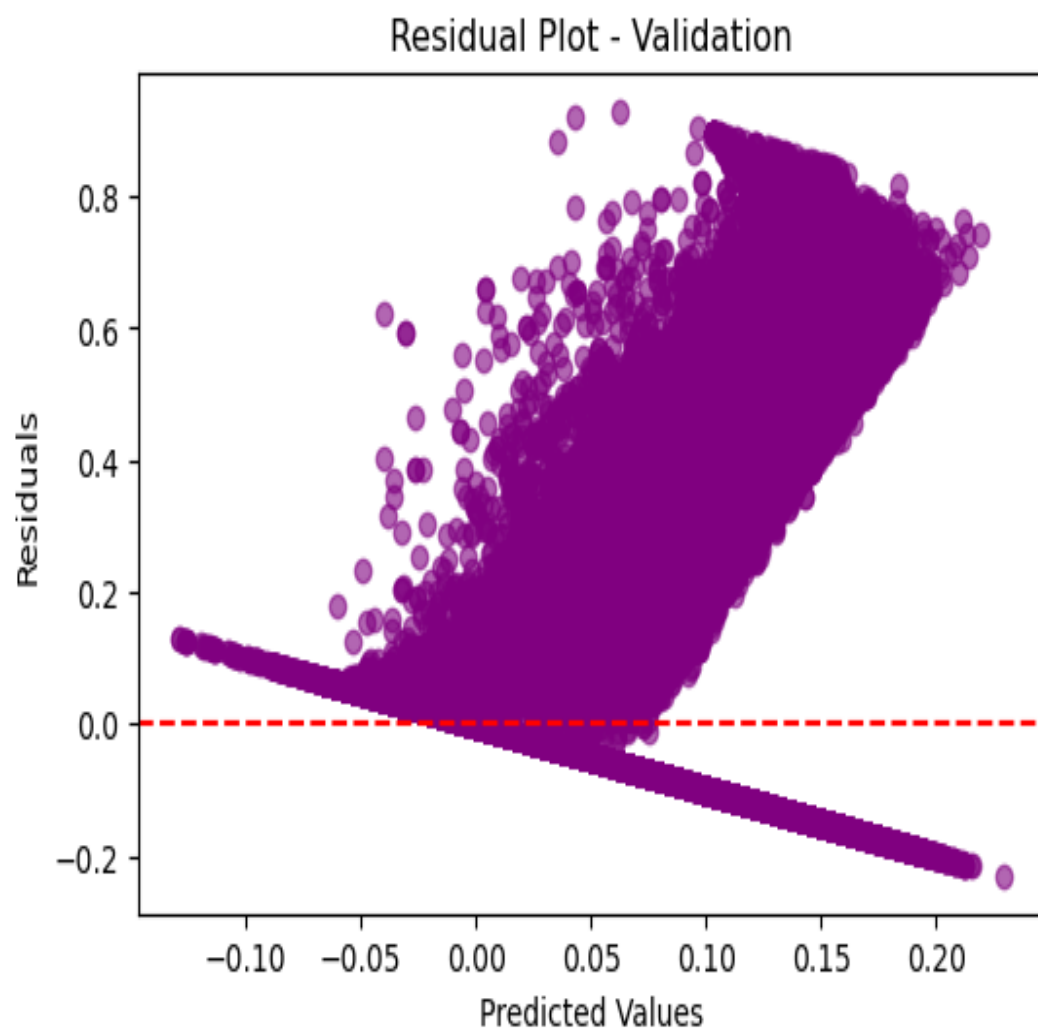
Prediction vs Plot - Validation



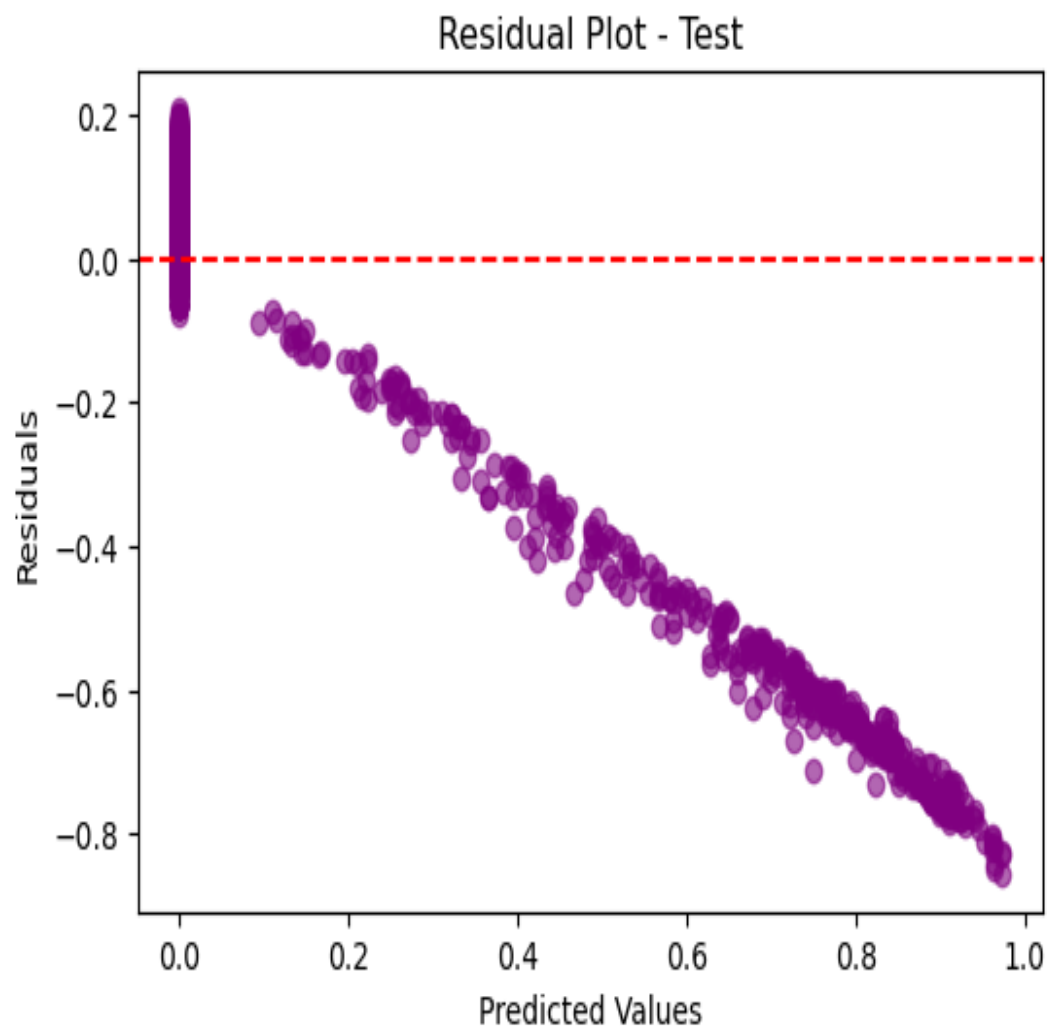
Prediction vs Plot - Test



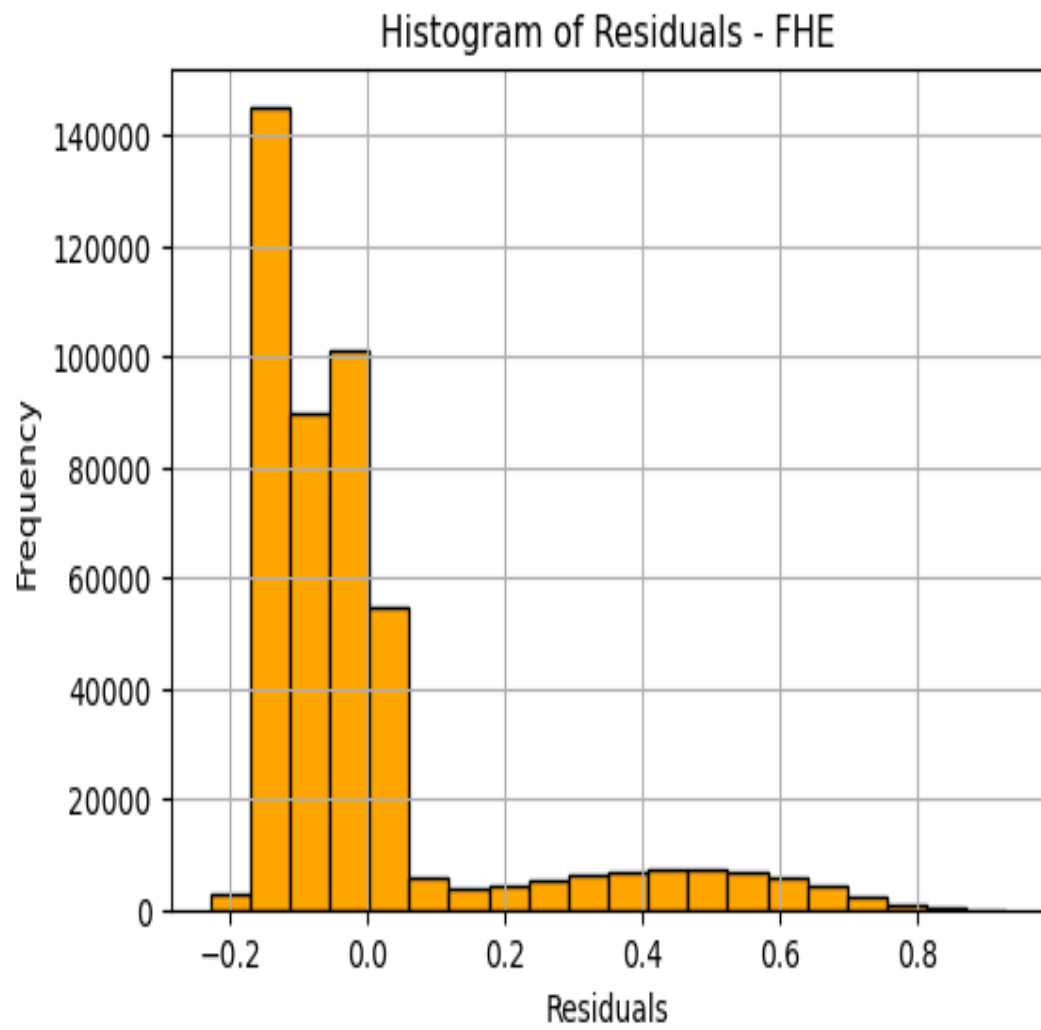
Residual Plot - Validation



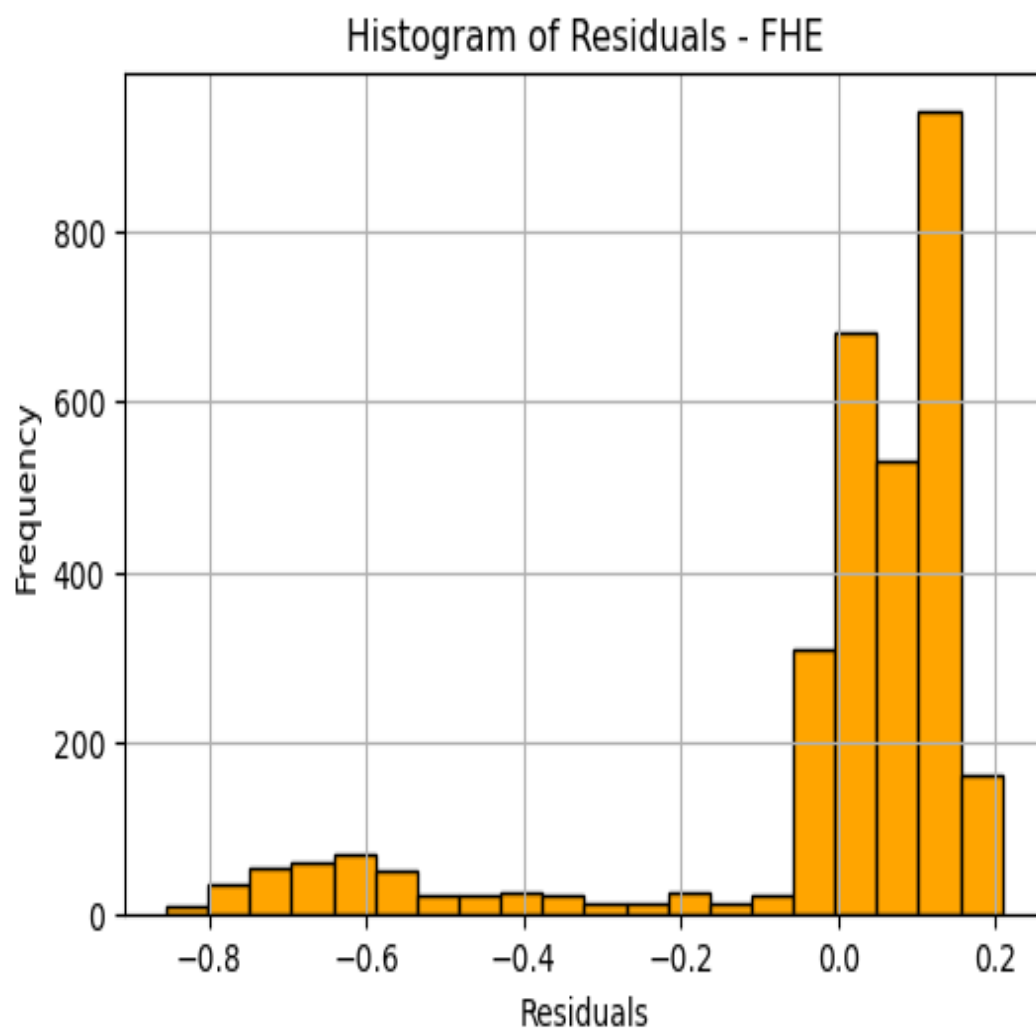
Residual Plot - Test



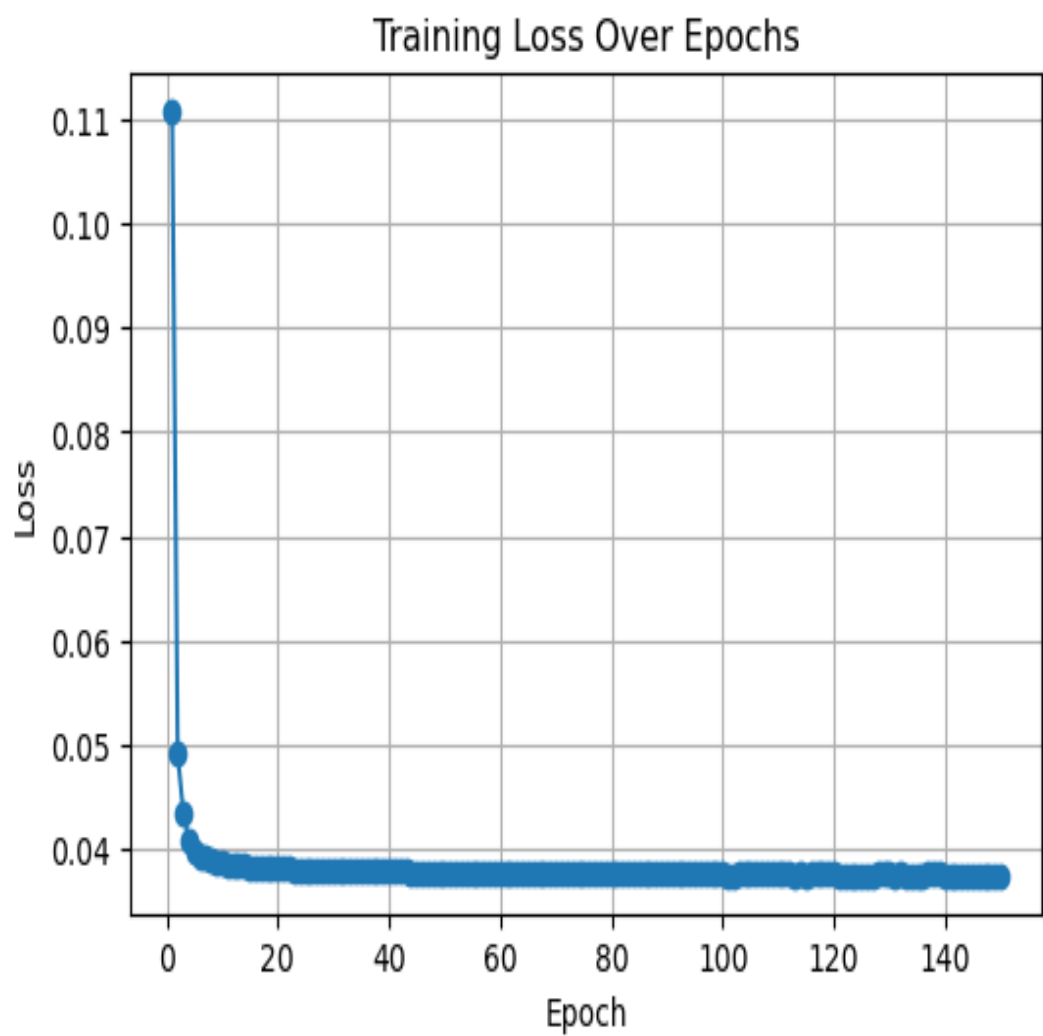
Residual Histogram - Validation



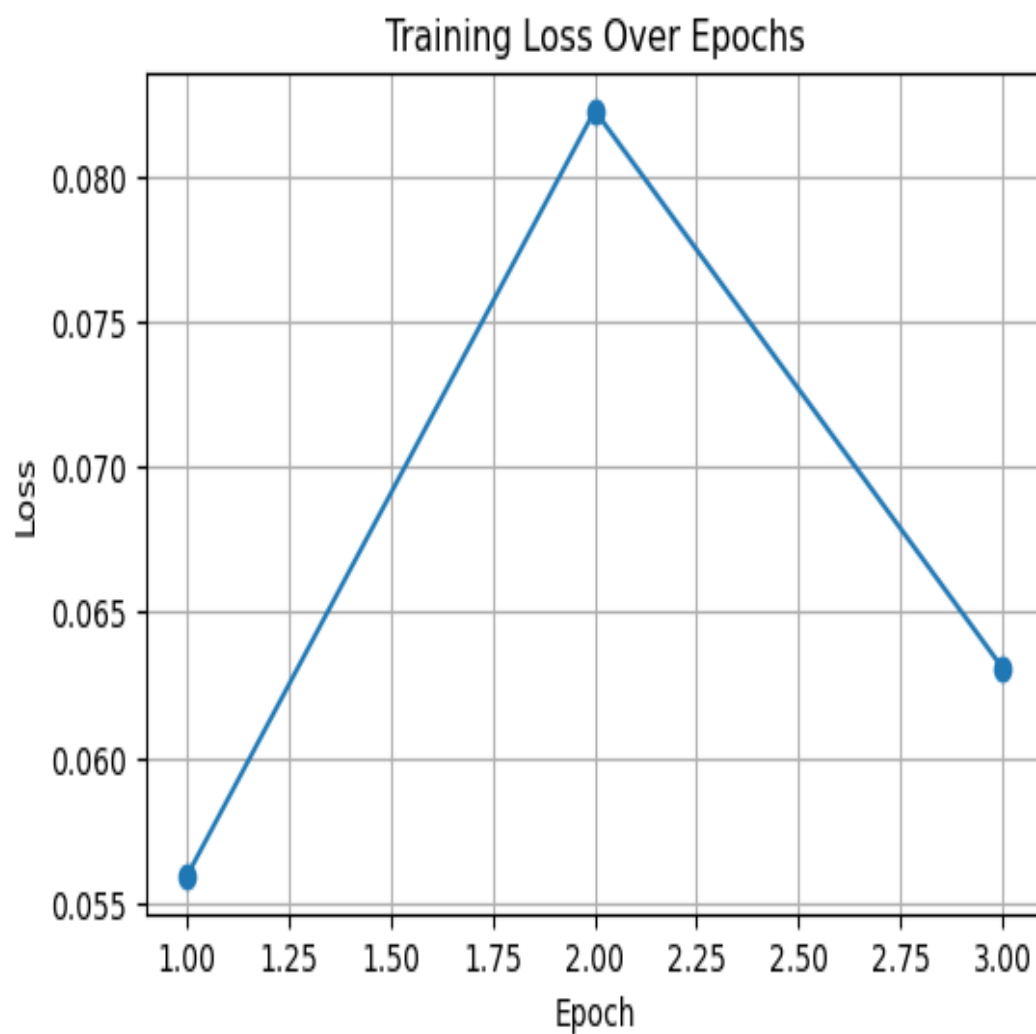
Residual Histogram - Test



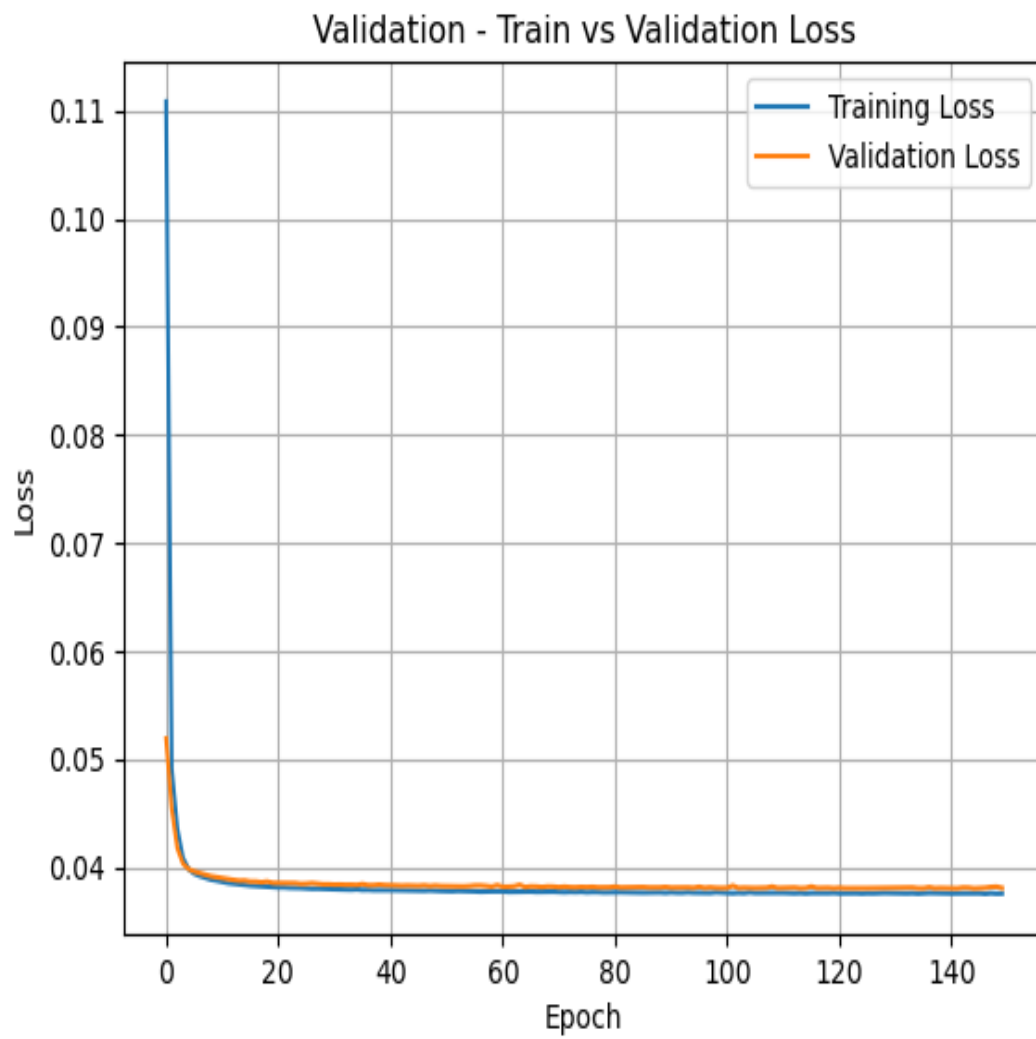
Training Loss - Validation



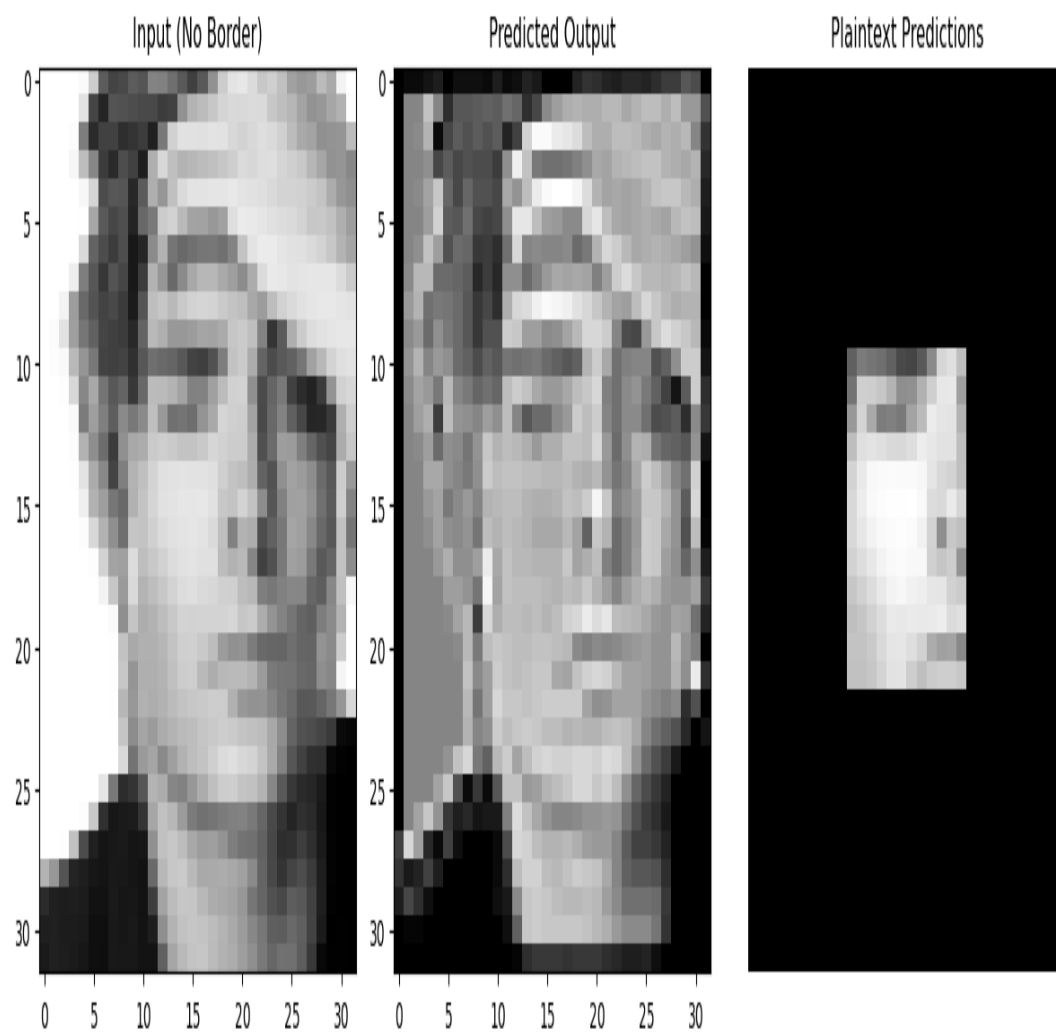
Training Loss - encrypted-fhe



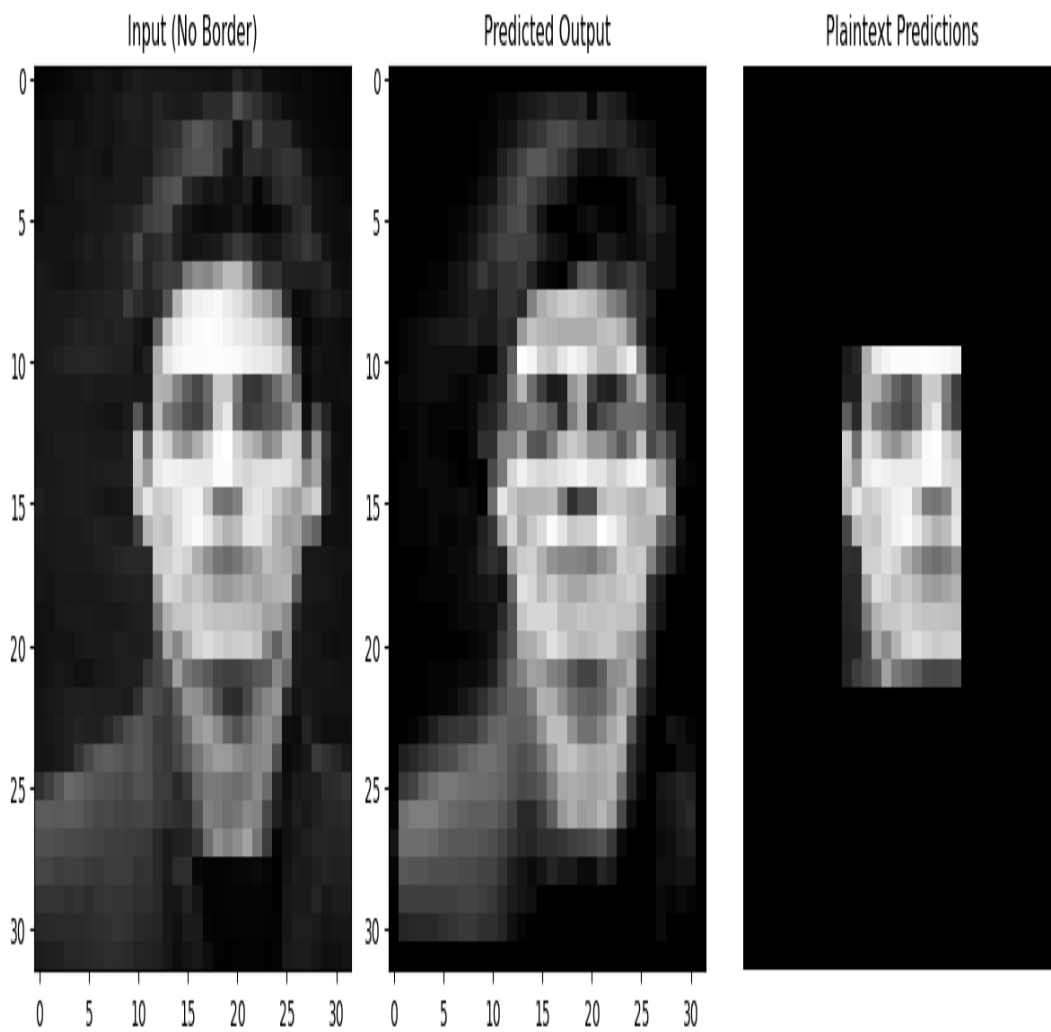
Train vs Validation Loss - Validation



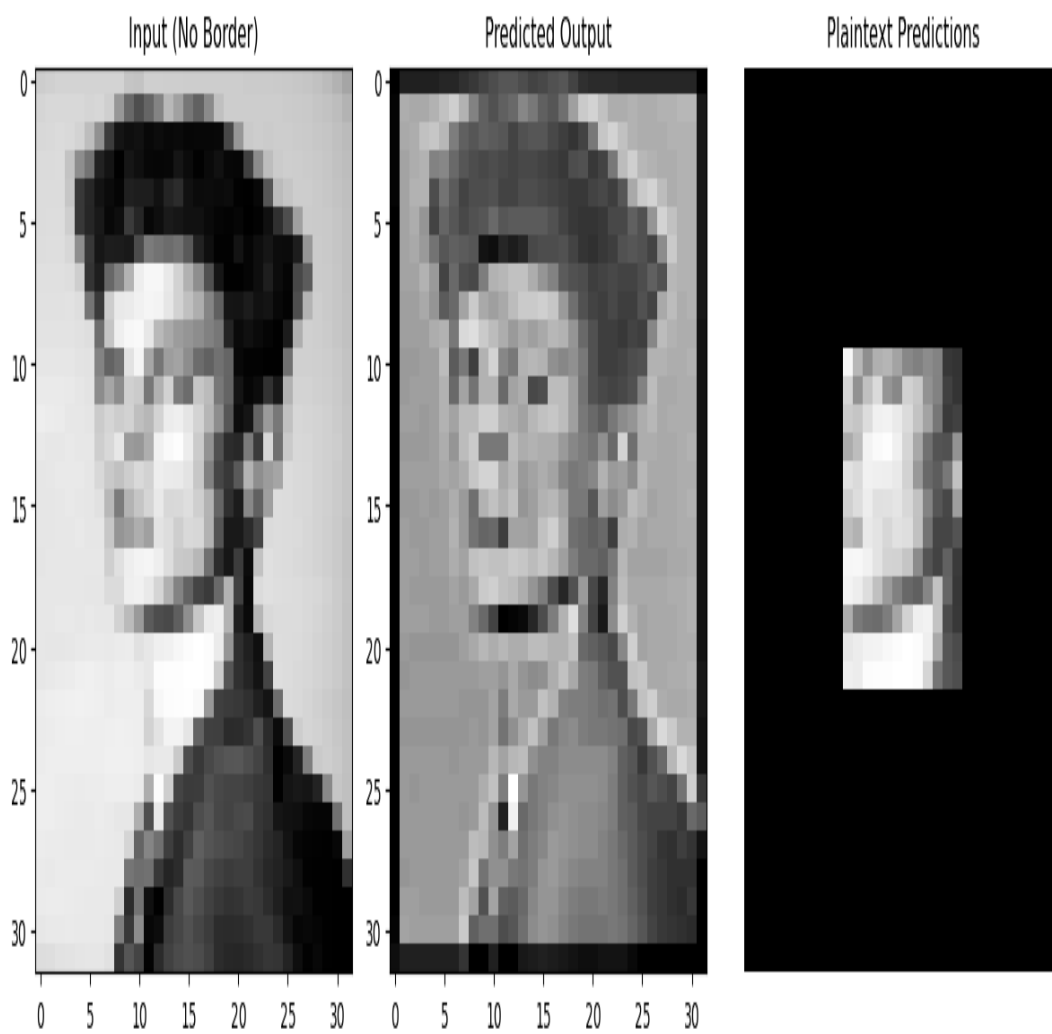
Plaintext: Model Prediction - Plaintext Predicted Image



Plaintext: Model Prediction - Plaintext Predicted Image



Plaintext: Model Prediction - Plaintext Predicted Image



Encrypted: Model Prediction - Encrypted Predicted Image

Encrypted Inference Predictions (Final_Model)

Input Image



Ground Truth



Predicted Output



Input Image



Ground Truth



Predicted Output



Input Image



Ground Truth



Encrypted Predictions



Runtime System Resource Usage

encrypted-fhe: OS: macOS-26.0-arm64-arm-64bit

encrypted-fhe: Python Version: 3.10.8

encrypted-fhe: Processor: arm

encrypted-fhe: CPU Usage (%): 5.0

encrypted-fhe: RAM Usage (Used GB): 6.62

encrypted-fhe: RAM Usage (Total GB): 18.0

encrypted-fhe: RAM Usage (%): 73.7

encrypted-fhe: GPU: Not Available

encrypted-fhe: CPU Usage (%): 1.8

encrypted-fhe: RAM Usage (Used GB): 6.57

encrypted-fhe: RAM Usage (Total GB): 18.0

encrypted-fhe: RAM Usage (%): 73.4

encrypted-fhe: GPU: Not Available

encrypted-fhe: CPU Usage (%): 1.6

encrypted-fhe: RAM Usage (Used GB): 6.56

encrypted-fhe: RAM Usage (Total GB): 18.0

encrypted-fhe: RAM Usage (%): 73.3

encrypted-fhe: GPU: Not Available

